Team Collaboration & Engineering Roster

Name	Primary Engineering Role	Milestone Contribution
Rachel Joy Baldo	Project Manager & Requirements Lead	Overall project coordination, PO4/M2CW reconciliation audit, requirements traceability.
Mark Angelo Aycardo	Software Architect & Lead Developer	Three-layer architectural enforcement, Strategy pattern implementation, operational contract verification.
Jian Bernard Quinton	Git Master & Systems Integrator	Codebase bundling, clean-room extraction testing, dependency isolation.
Issa Dumlao	Quality Assurance & Validation Lead	Automated test suite execution, precondition guard testing, failure invariant assertions.

1. Executive Summary & Baseline Identification

This release document accompanies the frozen release candidate of **ChallengeHub Prototype v0.2** for Module 2 Course Work (M2CW) evaluation. ChallengeHub is an event-driven virtual wellness and fitness challenge platform developed for the CPE106L-4 Software Design Laboratory.

The prototype consolidates individual Lesson 7 / M2LA2 implementations into a single, verified team baseline. It implements the vertical slice of **Record Activity (UC-05)**, dynamic **View Progress derivation (UC-04)**, in-memory precondition guards (FB-03), failure invariant preservation (FB-04), and atomic write-through flat-file persistence (FB-07).

Release Gate Deliverables Suite

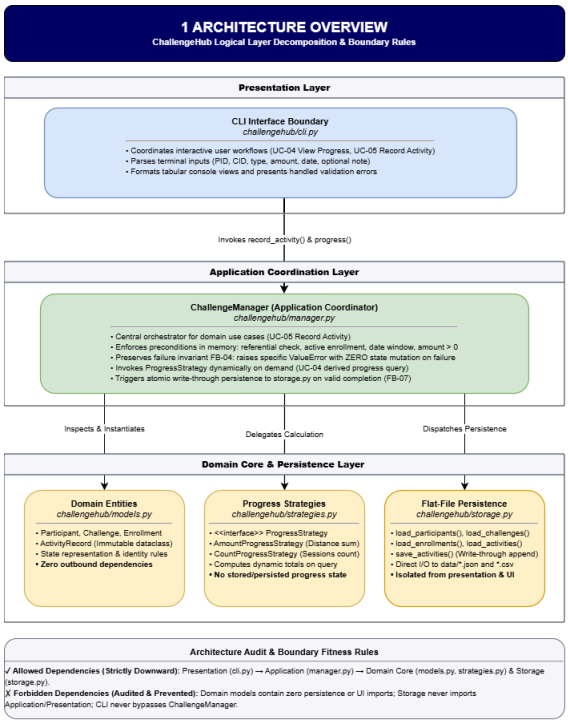
1. `Team01_M2CW_ReleaseCandidate.ipynb`: Complete 5-Act executable verification notebook.
2. `Team01_M2CW_Codebase.zip`: Standalone codebase archive (`challengehub/`, `tests/`, `data/`) verified via clean-room extraction.
3. `Team01_M2CW_DesignBaseline.drawio`: 5-page editable UML design workbook (Architecture, Class Model, Sequence Model, Strategy Pattern, Traceability Matrix).
4. `Team01_M2CW_ReleaseNotes.pdf`: This executive release memorandum.

Formal PO4 Data Reconciliation Statement

During M2CW release auditing, the team resolved a narrative discrepancy from early Project Output 4 drafting, where narrative text provisionally cited a 10.0 km goal target (75.0% completion). This was formally reconciled against the canonical `challenges.json` definition for Challenge `C001` (`Walk 20K`, `goal_target = 20.0 km`). With valid baseline activity entries totaling 7.5 km ($3.5\ \text{km} + 4.0\ \text{km}$), the true baseline completion benchmark is **37.5%** ($7.5/20.0$). The authoritative dataset was preserved without modification, fulfilling canonical baseline integrity.

2. Architecture & Pattern Decisions (ADR Summary)

ChallengeHub v0.2 enforces a strict **3-Layer Architectural Pattern** with unidirectional downward dependencies:



Architecture Decision Record (ADR)

- **Strategy Pattern (Selected & Implemented):** Distance-based challenges sum continuous quantities (`km`, `miles`), while session challenges tally discrete events (`session`, `count`). Hardcoding conditionals inside `ChallengeManager` violates the Open/Closed Principle. Extracting the `ProgressStrategy` interface with `AmountProgressStrategy` and `CountProgressStrategy` decouples metric algorithms from application workflow.
- **Repository Pattern (Simplified & Deferred):** Abstracting an interface over storage was deliberately deferred to Module IS. For Module 2, direct write-through file operations in `storage.py` maintain cohesion without unnecessary abstraction layers.
- **Factory & Observer Patterns (Rejected):** Uniform `ActivityRecord` schemas make standard dataclass instantiation concise, rendering a Factory pattern redundant. A single-threaded, synchronous CLI prototype has no requirement for asynchronous event broadcasting.

3. Reconciled Verification Matrix & Evidence

Verification Case	Target Requirement	Pre-Condition State	Stimulus / Operation	Actual Runtime Evidence	Status
Regression Smoke Suite	M2LA2 Compatibility	Clean repository checkout	`pytest -q` execution	3 passed in 0.06s; zero regressions against Lesson 7 baseline.	**PASS**
Normal Path Execution	FB-01, FB-02, UC-05	Count: 3 records Total: $7.5\ \text{km}$	`record_activity("P001", "C001", "walk", 2.0, 2026-08-10)`	Accepted `ActivityRecord`; count transitioned $3 \rightarrow 4$; progress derived to $9.5\ \text{km}$ (47.5%).	**PASS**
Boundary Valid Request	FB-01, UC-05	Count: 4 records Total: $9.5\ \text{km}$	`record_activity("P001", "C001", "walk", 0.01, 2026-08-01)`	Accepted edge amount ($0.01\ \text{km}$) on challenge start date; count transitioned $4 \rightarrow 5$; progress: $9.51\ \text{km}$.	**PASS**
Failure Invariant Guard	FB-03, FB-04, NFB-02	Count: 5 records Total: $9.51\ \text{km}$	`record_activity("P001", "C001", "walk", 0.0, 2026-08-10)`	Rejected with `ValueError: Amount must be positive`; **Count remained strictly** $5 \rightarrow 5$; **Total remained** $9.51\ \text{km}$ **(Zero mutation).**	**PASS**
Malformed Probe Isolation	FB-03, NFB-03	Baseline clean (3 rows)	Synthetic record `AR-MALFORMED-DEMO`	Flagged 3 concurrent defects (unknown PID, not enrolled, invalid amount); baseline preserved at 3 rows.	**PASS**
Persistence Reload Coherence	FB-07, NFB-03	Application memory cleared	Fresh `build_app(VERIFY_DATA)`	`activities.csv` contains 6 physical lines (1 header + 5 rows); reloaded manager restored count 5 and total $9.51\ \text{km}$.	**PASS**
Fresh-Extraction Test	Release Gate	Packaged codebase bundle	Extract ZIP to tempfile, run `pytest -q`	Extracted cleanly; pytest passed 3/3 in 0.02s; fresh `ChallengeManager` initialized baseline state ($7.5\ \text{km}$).	**PASS**

4. Known Limitations & Module IS Revision Targets

Documented Limitations in Prototype v0.2

1. **Flat-File Concurrency & Relational Constraints:** Persistence uses direct CSV/JSON serialization without transactional locks, atomic multi-file commits, or foreign-key cascade enforcement.
2. **Coupled Storage Boundary:** `ChallengeManager` directly references serialization functions in `storage.py` rather than coding to an abstracted persistence interface.
3. **Single-Slice Analytical Scope:** Progress summaries and Matplotlib visualizations target single `(participant_id, challenge_id)` pairs rather than multi-participant comparison matrices or leaderboards.

Specific Revision Targets for Module IS (Final Integration Project)

1. **Relational SQLite Migration:** Transition from flat files to an embedded SQLite database (`challengehub.db`) with foreign key constraints, unique indexing, and ACID transaction boundaries (FB-07, NFB-03).
2. **Formal Repository Pattern Implementation:** Implement an `ActivityRepository` interface to decouple application services from concrete persistence, enabling isolated unit testing with in-memory test doubles.
3. **Multi-Participant Comparative Dashboards:** Implement administrative reporting (UC-06 / FB-06) to compile aggregate leaderboards, completion rate distributions, and concurrent access tests.

5. Ethical AI-Assisted Use Disclosure & Accountability

Audit Category	Team Disclosure Record
What we asked AI to do	Consulted AI for drafting multi-layer pandas validation checks, Matplotlib progression visualizer templates, fresh-extraction pytest scripts, and 5-page Draw.io XML scaffolding.
Useful suggestion	Recommended separating data validation into non-mutating audit passes to prevent silent data dropping; suggested an automated extraction test using Python `tempfile` and `zipfile` to verify codebase independence.
How we verified it	Executed all unit tests and assertion cells inside isolated temporary directories; verified cumulative arithmetic independently ($3.5 + 4.0 = 7.5\ \text{km} \rightarrow 9.5\ \text{km} \rightarrow 9.51\ \text{km}$); confirmed via file inspection that zero state mutations occurred on rejected inputs.
What we corrected / rejected	Corrected date comparison logic to discrete calendar dates (`datetime.date`) to prevent timestamp clipping; rejected external save adapters because `ChallengeManager` persists atomically on write; rejected decorative Factory and Observer abstractions.
Final human team decision	Retained `PID="P001"` and `CID="C001"` across all notebook acts to preserve traceability with Act 1; maintained strict 3-layer architecture with dynamic Strategy-derived progress; froze the verified write-through baseline as the authoritative handoff for Module IS.

Baseline Release Freeze Status: Accepted and frozen as ChallengeHub Prototype v0.2 for Module IS handoff.

Team Sign-Off & Approval

- **Rachel Joy Baldo** (Project Manager & Requirements Lead)
- **Mark Angelo Aycardo** (Software Architect & Lead Developer)
- **Jian Bernard Quinton** (Git Master & Systems Integrator)
- **Issa Dumlao** (Quality Assurance & Validation Lead)

Release Status: `FROZEN BASELINE ACCEPTED FOR MODULE IS HANDOFF`